APPENDIX C
EVALUATION PSEUDOCODE

Rule-Based Evaluator (pseudocode)

```
# Input: itinerary text, user_budget, constraints
# Output: overall score S, dimension scores,
     accept/refine flag

weights = {budget:0.30, time:0.25, geo:0.20,
     diversity:0.15, constraints:0.10}

def evaluate(plan, budget, constraints):
   s_budget = score_budget(plan, budget) # 100 if
       <=90% budget, 0 if >130%
   s_time = score_time(plan)         # target: 8h
       activity + 2h travel/day
   s_geo  = score_geo(plan)          # ratio of
       route vs. 2-opt heuristic
   s_div  = score_diversity(plan)    # entropy over
       {culture, rec, dining,...}
   s_const = score_constraints(plan,constraints) #
       -50 hard, -20 soft violations

   scores = {budget:s_budget, time:s_time,
       geo:s_geo,
          diversity:s_div, constraints:s_const}

   S = sum(scores[d] * weights[d] for d in scores)

   # Threshold for acceptance
   accept = (S >= 75 and min(scores.values()) >= 60)
   return S, scores, accept
```

Fig. 2. Deterministic rule-based evaluator used to score itineraries across five dimensions.